Exp. No: 01	CLASSES AND OBJECTS
Date:	

PROGRAM :

```
import java.util.*;
class Box{
   double w,h,d;
   Box(){
      w=2;
      h=2;
      d=2;
   }
   Box(double a){
      w=a;
      h=2;
      d=3;
   }
   Box(double a,double b){
      w=a;
      h=b;
      d=3;
   }
   Box(double a,double b,double c){
      w=a;
      h=b;
      d=c;
   }
   double Volume(){
      return w*h*d;
   }
```

```
}
class Vol_Box{
        public static void main(String args[]) {
        Scanner s=new Scanner(System.in);
        System.out.println("\t Classes and Objects");
        double a1,b1,b2,c1,c2,c3;
        System.out.println("Enter One value :");
        a1=s.nextDouble();
        System.out.println("Enter Two values :");
        b1=s.nextDouble();
        b2=s.nextDouble();
        System.out.println("Enter Three values :");
        c1=s.nextDouble();
        c2=s.nextDouble();
        c3=s.nextDouble();
        Box box=new Box();
        Box box1=new Box(a1);
        Box box2=new Box(b1,b2);
        Box box3=new Box(c1,c2,c3);
        System.out.println("The Volume of Box1 is :"+box.Volume());
        System.out.println("The Volume of Box2 is :"+box1.Volume());
        System.out.println("The Volume of Box3 is :"+box2.Volume());
        System.out.println("The Volume of Box4 is :"+box3.Volume());
    }
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java Vol_Box
         Classes and Objects
Enter One value :
2
Enter Two values :
2
3
Enter Three values :
4
5
6
The Volume of Box1 is :8.0
The Volume of Box2 is :12.0
The Volume of Box3 is :18.0
The Volume of Box4 is :120.0
```

Exp. No : 02	IMPLEMENTING CLASSES
Date:	

PROGRAM :

```
import java.util.*;
class Area_Shapes{
   double pi=3.14;
   void area(int a){
      System.out.println("\nThe Area of the Square is :"+a*a);
   }
   void area(int a,int b){
      System.out.println("The Area of the Rectangle is :"+a*b);
   }
   double area(double r){
      return pi*r*r;
   }
}
class Shapes{
   public static void main(String args[]){
   System.out.println("\t Implementing Classes");
   Scanner s=new Scanner(System.in);
   double c,area_cir;
   int sq,l,b;
   Area_Shapes ar=new Area_Shapes();
   System.out.print("\nEnter the Side for Square :");
   sq=s.nextInt();
   System.out.print("Enter the Length of the Rectangle :");
   l=s.nextInt();
   System.out.print("Enter the Breadth of the Rectangle :");
   b=s.nextInt();
```

```
    System.out.print("Enter the Radius of the Circle :");
    c=s.nextDouble();
    ar.area(sq);
    ar.area(l,b);
    area_cir=ar.area(c);
    System.out.println("The Area of the Circle is :"+area_cir);
    }
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java Shapes
         Implementing Classes

Enter the Side for Square :2
Enter the Length of the Rectangle :3
Enter the Breadth of the Rectangle :4
Enter the Radius of the Circle :4

The Area of the Square is :4
The Area of the Rectangle is :12
The Area of the Circle is :50.24
```

Exp. No: 03	STRINGS
Date:	

PROGRAM :

```
import java.util.*;
class String_func{
   public static void main(String args[]) {
      Scanner s=new Scanner(System.in);
      System.out.print("Enter the String :");
      String str=s.nextLine();
      System.out.print("Enter the string to check the given string is starts with it :");
      String a1=s.nextLine();
      System.out.print("Enter the string to check the given string is Ends with it :");
      String a2=s.nextLine();
      System.out.print("Enter the string to find position :");
      String p1=s.nextLine();
      System.out.print("Enter the position to find character :");
      int p2=s.nextInt();
      s.nextLine();
      System.out.println("\n");
      System.out.println("The length of the string is :"+str.length());
      System.out.println("The Upper case is :"+str.toUpperCase());
      System.out.println("The Lower case is :"+str.toLowerCase());
      System.out.println("The string is starts with the \""+a1+"\"
is :"+str.startsWith(a1));
      System.out.println("The string is ends with the \""+a2+"\"
is :"+str.endsWith(a2));
      System.out.println(p1+" position is :"+str.indexOf(p1));
      System.out.println("The character in "+p2+" is :"+str.charAt(p2));
      System.out.println("The substring is :"+str.substring(1,7));
```

```
        // String Buffer
        System.out.println("\n----- StringBuffer Functions -----");
        StringBuffer str2=new StringBuffer(str);
        System.out.print("Enter the String to append :");
        String ap=s.nextLine();
        str2.append(ap);
        System.out.println("After append :"+str2);
        System.out.println("\n");
        System.out.print("Enter the String to Insert :");
        String i1=s.nextLine();
        System.out.print("Enter the position to Insert :");
        int i2=s.nextInt();
        str2.insert(i2,i1);
        System.out.println("After Insert :"+str2);
        System.out.println("\n");
        System.out.print("Enter the position to delete a character :");
        int d1=s.nextInt();
        str2.deleteCharAt(d1);
        System.out.println("After delete : "+str2);
        System.out.println("\n");
        System.out.println("Reverse string :"+str2.reverse());
    }
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java String_func
Enter the String :Hi Iam Ragupathi
Enter the string to check the given string is starts with it :H
Enter the string to check the given string is Ends with it :r
Enter the string to find position :R
Enter the position to find character :3


The length of the string is :16
The Upper case is :HI IAM RAGUPATHI
The Lower case is :hi iam ragupathi
The string is starts with the "H" is :true
The string is ends with the "r" is :false
R position is :7
The character in 3 is :I
The substring is :i Iam

----- StringBuffer Functions -----
Enter the String to append : stud
After append :Hi Iam Ragupathi stud


Enter the String to Insert : Hello
Enter the position to Insert :2
After Insert :Hi Hello Iam Ragupathi stud


Enter the position to delete a character :1
After delete : H Hello Iam Ragupathi stud


Reverse string :duts ihtapugaR maI olleH H
```

Exp. No: 04	COLLECTION
Date:	

PROGRAM :

```
import java.util.*;
class Collection{
   public static void main(String args[]) {
      Scanner s=new Scanner(System.in);
      ArrayList<Integer> al=new ArrayList<>();
      System.err.println("Enter Five Numbers :");
      for(int i=1;i<=5;i++){
         System.out.print("Enter the Element "+i+" :");
         al.add(s.nextInt());
      }
      int sum=0;
      for (int a:al){
         sum+=a;
      }
      System.out.println("The Sum of the Elements is :"+sum);
      LinkedList<String> l=new LinkedList<>();
      HashSet<String> hs=new HashSet<>();
      TreeSet<String> ts=new TreeSet<>();
      System.out.println("\nEnter Five Letters :");
      s.nextLine();
      for(int i=1;i<=5;i++){
         System.out.print("Enter the Letter "+i+" :");
         String str=s.nextLine();
         l.add(str);
         hs.add(str);
         ts.add(str);
```

```
        }
        System.out.println("\nThe Linked List Elements are :"+l);
        System.out.print("Enter the Element to remove :");
        String s1=s.nextLine();
        System.out.print("Enter the Position to remove :");
        int s2=s.nextInt();
        System.out.println("Before Remove :"+l);
        l.remove(s1);
        l.remove(s2);
        System.out.println("After Remove :"+l);
        System.out.println("\nThe HashSet(Unordered) elements are :"+hs);
        System.out.println("\nThe Treeset(Ordered) Elements are :"+ts);
    }
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java Collection
Enter Five Numbers :
Enter the Element 1 :2
Enter the Element 2 :3
Enter the Element 3 :4
Enter the Element 4 :5
Enter the Element 5 :6
The Sum of the Elements is :20

Enter Five Letters :
Enter the Letter 1 :A
Enter the Letter 2 :S
Enter the Letter 3 :D
Enter the Letter 4 :F
Enter the Letter 5 :G

The Linked List Elements are :[A, S, D, F, G]
Enter the Element to remove :S
Enter the Position to remove :2
Before Remove :[A, S, D, F, G]
After Remove :[A, D, G]

The HashSet(Unordered) elements are :[A, S, D, F, G]

The Treeset(Ordered) Elements are :[A, D, F, G, S]
```

Exp. No: 05	DATE AND CALENDAR
Date:	

PROGRAM :

```
import java.util.*;
class Date_Time{
   public static void main(String[] args) {
      Scanner s=new Scanner(System.in);
      Calendar cal=Calendar.getInstance();
      Date dt=new Date();
      System.out.println("\t Date and Time" );
      System.out.println("\nThe Current Date,Time and Timezone in a System is :
      “+dt);
      System.out.println("The Current Time in Milli Second is :"+dt.getTime());
      String months[]=
       {"Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Oct","Sep","Nov","Dec"};
      System.out.print("\nDate : ");
      System.out.print(months[cal.get(Calendar.MONTH)]);
      System.out.print(" " + cal.get(Calendar.DATE));
      System.out.print(", " + cal.get(Calendar.YEAR));

      System.out.print("\nTime : ");
      System.out.print(cal.get(Calendar.HOUR));
      System.out.print(" : "+cal.get(Calendar.MINUTE));
      System.out.print(" : "+cal.get(Calendar.SECOND));

      System.out.println("\n");
      System.out.print("Enter the Hour :");
      int h=s.nextInt();
      System.out.print("Enter the Minutes :");
```

```
        int m=s.nextInt();
        System.out.print("Enter the Seconds :");
        int sc=s.nextInt();
        cal.set(Calendar.HOUR,h);
        cal.set(Calendar.MINUTE,m);
        cal.set(Calendar.SECOND,sc);
        System.out.print("\nUpdated Time :");
        System.out.print(cal.get(Calendar.HOUR));
        System.out.print(" : "+cal.get(Calendar.MINUTE));
        System.out.print(" : "+cal.get(Calendar.SECOND)+"\n");
    }
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java Date_Time
         Date and Time

The Current Date,Time
 and Timezone in a System is : Sat Sep 26 14:52:30 IST 2026
The Current Time in Milli Second is :1790414550901

Date : Oct 26, 2026
Time : 2 : 52 : 30

Enter the Hour :10
Enter the Minutes :20
Enter the Seconds :15

Updated Time :10 : 20 : 15
```

Exp. No: 06	PACKAGES
Date:	

PROGRAM :

Account.java :

```
package Bank;
public class Account {
   String name;
   int balance;
   public void createAccount(String name, int initialAmount) {
      this.name = name;
      this.balance = initialAmount;
   }
   public int credit(int amount){
      balance+=amount;
      System.out.println("The Amount Credited Successfully");
      return balance;
   }
   public int withdraw(int amount){
      if(amount<=balance){
         balance-=amount;
         System.out.println("The Amount Withdrawn Successfully");
      }
      else{
         System.out.println("Your balance is Low");
      }
      return balance;
      }
   public void disp_balance(){
      System.out.println("Name :"+name);
```

```
        System.out.println("Balance :"+balance);
    }
}
```

User.java :

```
import Bank.Account;
import java.util.*;
class User{
    public static void main(String[] args) {
        Scanner s= new Scanner(System.in);
        Account ac=new Account();
        System.out.print("\nEnter the Name :");
        String name=s.nextLine();
        System.out.print("\nEnter the Initial Amount :");
        int amount=s.nextInt();
        s.nextLine();
        ac.createAccount(name, amount);
        System.out.println("""
                1)Credit
                2)withdraw
                3)Balance
                4)Exit
                """);
        while(true){
        System.out.print("\nEnter What you want to do :");
        String msg=s.nextLine().toLowerCase();
        if("credit".equals(msg)){
```

```
            System.out.print("\nEnter the amount :");
            int amt=s.nextInt();
            ac.credit(amt);
            s.nextLine();
        }
        else if("withdraw".equals(msg)){
            System.out.print("\nEnter the amount :");
            int amt=s.nextInt();
            ac.withdraw(amt);
            s.nextLine();
        }
        else if("balance".equals(msg)){
            ac.disp_balance();
        }
        else if("exit".equals(msg)){
            break;
        }
        else{
            System.out.println("Invalid Action !");
        }
        }
    }
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java User

Enter the Name :Ragupathi

Enter the Initial Amount :1500
1)Credit
2)withdraw
3)Balance
4)Exit


Enter What you want to do :credit

Enter the amount :1000
The Amount Credited Successfully

Enter What you want to do :withdraw

Enter the amount :500
The Amount Withdrawn Successfully

Enter What you want to do :balance
Name :Ragupathi
Balance :2000

Enter What you want to do :exit
```

Exp. No: 07	EXCEPTION HANDLING
Date:	

PROGRAM :

```
import java.util.*;
class Exception_Hand {
    public static void main(String[] args) {
        int a = args.length;
        Scanner s = new Scanner(System.in);
        System.out.println("Value of a is : " + a);
        try {
            if (a == 0) {
                int b = 24;
                a = b / (b - b);
            }
            if (a == 1) {
                int A[] = {};
                A[16] = 20;
            }
            if (a == 2) {
                System.out.print("Enter Array size: ");
                int n = s.nextInt();
                int B[]= new int[n];
                B[17] = 15;
            }
        } catch (ArithmeticException e) {
            System.out.println("Do not use zero as a divisor. " + e);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Array size exceeds. " + e);
        }
```

```
        catch (NegativeArraySizeException e) {
            System.out.println("Array size cannot be negative value. " + e);
        } finally {
            System.out.println("Exceptions Completed.");
        }
    }
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java Exception_Hand
Value of a is : 0
Do not use zero as a divisor. java.lang.ArithmeticException: / by zero
Exceptions Completed.
ragupathi@ragu-2k05:~$ java Exception_Hand 3
Value of a is : 1
Array size exceeds. java.lang.ArrayIndexOutOfBoundsException: Index 16 out of bounds for length 0
Exceptions Completed.
ragupathi@ragu-2k05:~$ java Exception_Hand 3 4
Value of a is : 2
Enter Array size: -5
Array size cannot be negative value. java.lang.NegativeArraySizeException: -5
Exceptions Completed.
```

Exp. No: 08	THREADS
Date:	

PROGRAM :

```
class Q {
   int n;
   boolean valueSet = false;
   synchronized int get() {
      if (!valueSet)
         try {
            wait();
         } catch (InterruptedException e) {
            System.out.println("InterruptedException caught");
         }
      System.out.println("Got: " + n);
      valueSet = false;
      notify();
      return n;
   }
   synchronized void put(int n) {
      if (valueSet)
         try {
            wait();
         } catch (InterruptedException e) {
            System.out.println("InterruptedException caught");
         }
      this.n = n;
      valueSet = true;
      System.out.println("Put: " + n);
      notify();
```

```
    }
}
class Producer implements Runnable {
    Q q;
    Producer(Q q) {
        this.q = q;
        new Thread(this, "Producer").start();
    }
    public void run() {
        int i = 1;
        while (i <= 10) {
            q.put(i++);
        }
    }
}
class Consumer implements Runnable {
    Q q;
    Consumer(Q q) {
        this.q = q;
        new Thread(this, "Consumer").start();
    }
    public void run() {
        while (true) {
            q.get();
        }
    }
}
```

```
class PC {
    public static void main(String args[]) {
        System.out.println("Press Control-C to stop.");
        Q q = new Q();
        new Producer(q);
        new Consumer(q);
    }
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java PC
Press Control-C to stop.
Put: 1
Got: 1
Put: 2
Got: 2
Put: 3
Got: 3
Put: 4
Got: 4
Put: 5
Got: 5
Put: 6
Got: 6
Put: 7
Got: 7
Put: 8
Got: 8
Put: 9
Got: 9
Put: 10
Got: 10
```

Exp. No: 09	JDBC
Date:	

PROGRAM :

Database code :

```
mysql> use ragu
Database changed
mysql> create table student(Id int,name varchar(20),age int,class varchar(20));
Query OK, 0 rows affected (0.10 sec)

mysql> insert into student values(01,"Ragu",20,"12 th"),(02,"Jeffy",21,"11 th"),(03,"Walter",22,"10 th");
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select*from student;
+------+--------+------+-------+
| Id   | name   | age  | class |
+------+--------+------+-------+
|    1 | Ragu   |   20 | 12 th |
|    2 | Jeffy  |   21 | 11 th |
|    3 | Walter |   22 | 10 th |
+------+--------+------+-------+
3 rows in set (0.00 sec)
```

Student.java :

```
import java.sql.*;

public class Student {

    public static void main(String args[]) {

        try {

            // Load MySQL JDBC Driver

            Class.forName("com.mysql.cj.jdbc.Driver");


            // Database connection

            String url = "jdbc:mysql://localhost:3306/ragu";

            String user = "root";

            String password = "ragu@123";
```

```
        Connection con = DriverManager.getConnection(url, user, password);
        System.out.println("Created DB Connection...");
        Statement stmt = con.createStatement();

        // Execute Query
        ResultSet rs = stmt.executeQuery("SELECT Id, name, class FROM student");

        // Print Column Names
        ResultSetMetaData rsmd = rs.getMetaData();
        int count = rsmd.getColumnCount();
        for (int i = 1; i <= count; i++) {
            System.out.print(rsmd.getColumnName(i) + "\t");
        }
        System.out.println();

        // Print Records
        while (rs.next()) {
            System.out.print(rs.getInt("id") + "\t");
            System.out.print(rs.getString("name") + "\t");
            System.out.print(rs.getString("class") + "\t");
            System.out.println();
        }
        // Close Resources
        rs.close();
        stmt.close();
        con.close();
    } catch (ClassNotFoundException e) {
```

```
            System.out.println("JDBC Driver Not Found!");
            e.printStackTrace();
        } catch (SQLException e) {
            System.out.println("Database Connection Error!");
            e.printStackTrace();
        }
    }
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java -cp ".:lib/mysql-connector.jar" Student
Created DB Connection...
Id      name    class
1       Ragu    12 th
2       Jeffy   11 th
3       Walter  10 th
```

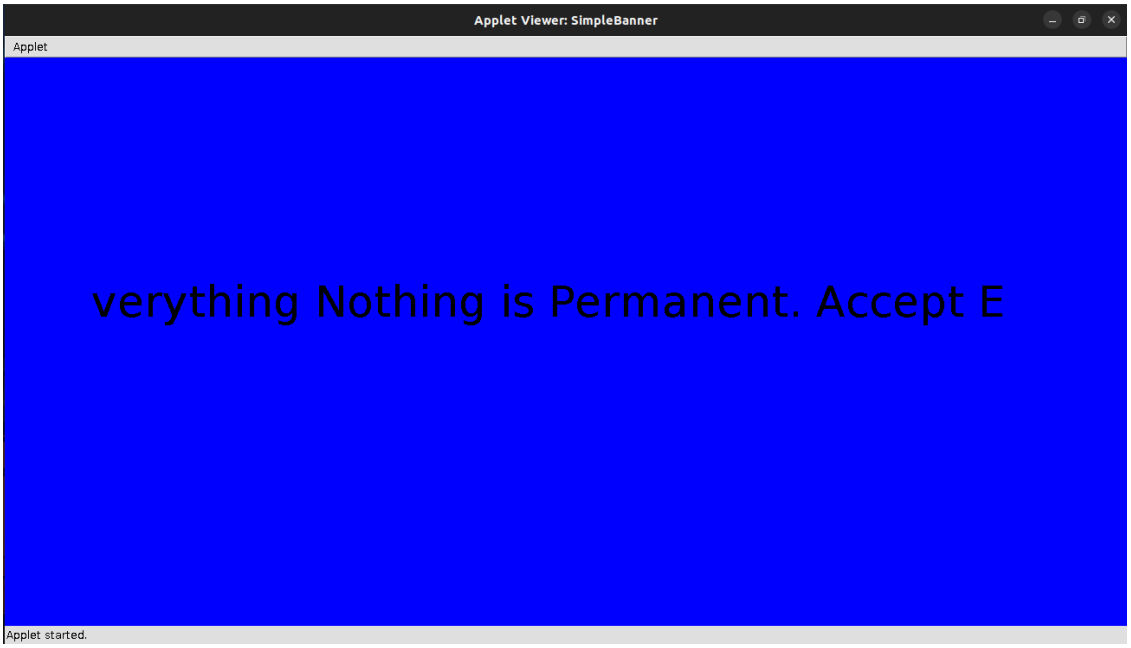
Exp. No: 10	APPLETS
Date:	

PROGRAM :

```
import java.awt.*;
import java.applet.*;
/*
<applet code="SimpleBanner" width="700" height="50"> </applet>
*/
public class SimpleBanner extends Applet implements Runnable {
    String msg = " Accept Everything Nothing is Permanent.";
    Thread t = null;
    int state;
    boolean stopFlag;
    // Set colors and initialize thread
    public void init() {
        setBackground(Color.blue);
        setForeground(Color.white);
        setFont(new Font("Dialog", Font.PLAIN, 50));
    }
    // Start thread
    public void start() {
        t = new Thread(this);
        stopFlag = false;
        t.start();
    }
    // Entry point for the thread that runs the banner
    public void run() {
        char ch;
```

```
        // Display banner
        for (;;) {
            try {
                repaint();
                Thread.sleep(250);
                ch = msg.charAt(0);
                msg = msg.substring(1, msg.length());
                msg += ch;
                if (stopFlag)
                    break;
            } catch (InterruptedException e) {
            }
        }
    }
    // Pause the banner
    public void stop() {
        stopFlag = true;
        t = null;
    }
    // Display the banner
    public void paint(Graphics g) {
        g.drawString(msg, 100, 300);
    }
}
```

OUTPUT :


Applet Viewer: SimpleBanner
Applet
verything Nothing is Permanent. Accept E
Applet started.

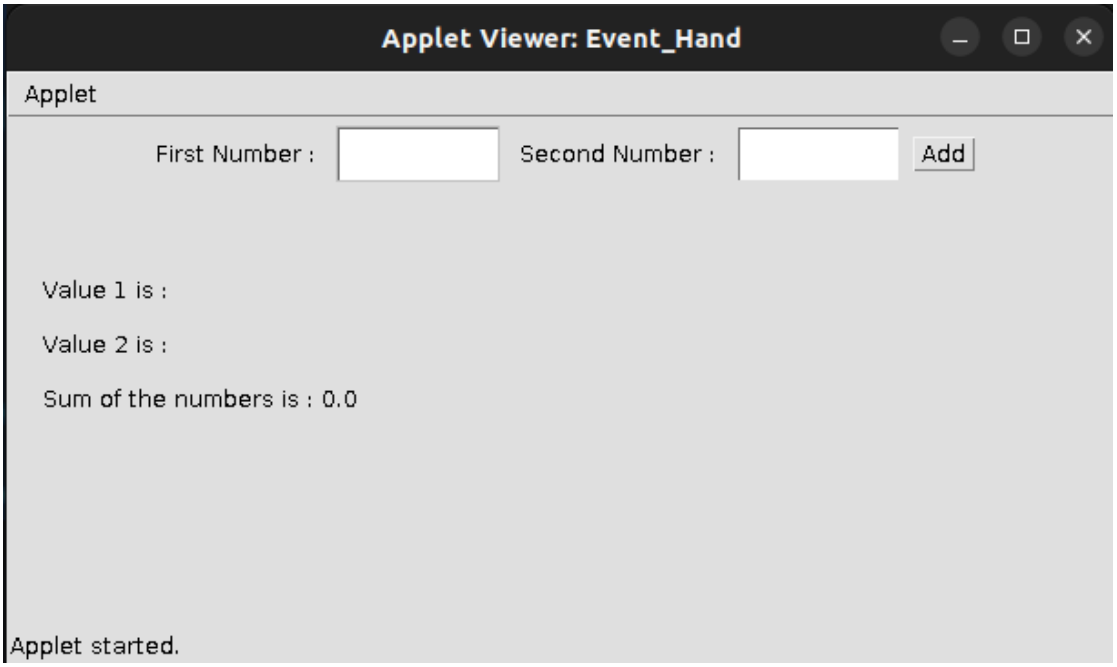
Exp. No: 11	EVENT HANDLING
Date:	

PROGRAM :

```
import java.applet.*;
import java.awt.*;
import java.awt.event.*;
/*
<applet code="Event_Hand" width="1200" height="1200">
</applet>
*/
public class Event_Hand extends Applet implements ActionListener {
   TextField a, b;
   double sum;
   Button ab;
   public void init() {
      Label value1 = new Label("First Number : ", Label.RIGHT);
      Label value2 = new Label("Second Number : ", Label.RIGHT);
      a = new TextField(10);
      b = new TextField(10);
      ab = new Button("Add");
      add(value1);
      add(a);
      add(value2);
      add(b);
      add(ab);
      ab.addActionListener(this);
   }
   public void actionPerformed(ActionEvent ae) {
```

```
        if (ae.getSource() == ab) {
            try {
                int num1 = Integer.parseInt(a.getText());
                int num2 = Integer.parseInt(b.getText());
                sum = num1 + num2;
            } catch (NumberFormatException e) {
                sum = 0;
            }
            repaint();
        }
    }
    public void paint(Graphics g) {
        g.drawString("Value 1 is : " + a.getText(), 20, 100);
        g.drawString("Value 2 is : " + b.getText(), 20, 130);
        g.drawString("Sum of the numbers is : " + sum, 20, 160);
    }
}
```

OUTPUT :


Applet Viewer: Event_Hand
Applet
First Number :
Second Number :
Add
Value 1 is :
Value 2 is :
Sum of the numbers is : 0.0
Applet started.

Exp. No: 12	SIMPLE WEB APPLICATIONS
Date:	

PROGRAM :

Index.html :

```
<html>
<head>
   <title>Voting Eligibility Check</title>
</head>
<body>
<h2>Check Voting Eligibility</h2>
<form action="vote" method="post">
   <label for="name">Name:</label>
   <input type="text" id="name" name="name" required>
   <br><br>
   <label for="age">Age:</label>
   <input type="number" id="age" name="age" required>
   <br><br>
   <input type="submit" value="Check Eligibility">
</form>
</body>
</html>
```

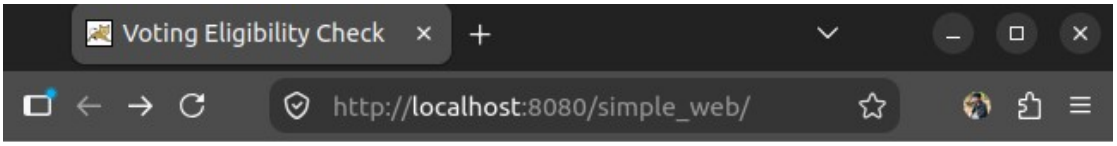
vote.java :

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
```

```
import javax.servlet.http.HttpServletResponse;


@WebServlet("/vote")
public class vote extends HttpServlet {
   @Override
   protected void doPost(HttpServletRequest request,
         HttpServletResponse response)
         throws ServletException, IOException {
      String name = request.getParameter("name");
      int age = Integer.parseInt(request.getParameter("age"));
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      out.println("<html><body>");
      if (age < 18) {
         out.println("<h2>" + name + " is not eligible to vote.</h2>");
      } else {
         out.println("<h2>" + name + " is eligible to vote.</h2>");
      }
      out.println("</body></html>");
      out.close();
   }
}
```

OUTPUT :

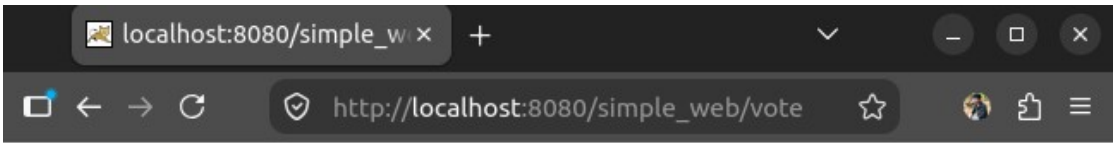


Check Voting Eligibility

Name: Ragupathi

Age: 19

Check Eligibility



Ragupathi is eligible to vote.

Exp. No: 13	USING SESSIONS AND COOKIES
Date:	

PROGRAM :

Index.html :

```
<html>
<head>
   <title>Login Page</title>
</head>
<body>
<h2>Login</h2>
<form action="LoginServlet" method="POST">
   Name:
   <input type="text" name="username">
   <br><br>
   <input type="submit" value="Submit">
</form>
</body>
</html>
```

LoginServlet.java :

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {
   protected void doPost(HttpServletRequest request,
                         HttpServletResponse response)
```

```
            throws ServletException, IOException {
        response.setContentType("text/html");
        String username = request.getParameter("username");
        // Create Session
        HttpSession session = request.getSession();
        session.setAttribute("username", username);
        // Create Cookie
        Cookie userCookie = new Cookie("username", username);
        userCookie.setMaxAge(60 * 60 * 24); // 1 day
        response.addCookie(userCookie);
        response.sendRedirect("WelcomeServlet");
    }
}
```

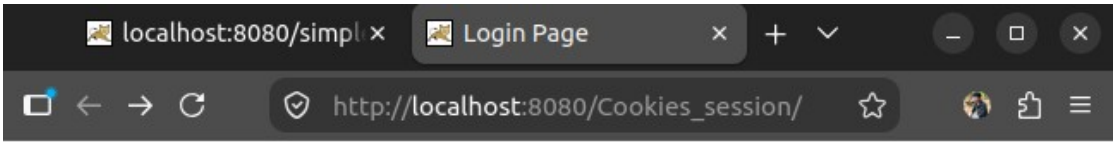
WelcomeServlet.java :

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/WelcomeServlet")
public class WelcomeServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
```

```
        HttpSession session = request.getSession(false);
        String username = null;
        if (session != null) {
            username = (String) session.getAttribute("username");
        }
        if (username == null) {
            Cookie[] cookies = request.getCookies();
            if (cookies != null) {
                for (Cookie cookie : cookies) {
                    if (cookie.getName().equals("username")) {
                        username = cookie.getValue();
                        break;
                    }
                }
            }
        }
        if (username != null) {
            out.println("<h2>Welcome, " + username + "!</h2>");
        } else {
            out.println("<h2>No user found. Please log in.</h2>");
            out.println("<a href='index.html'>Login</a>");
        }
    }
}
```

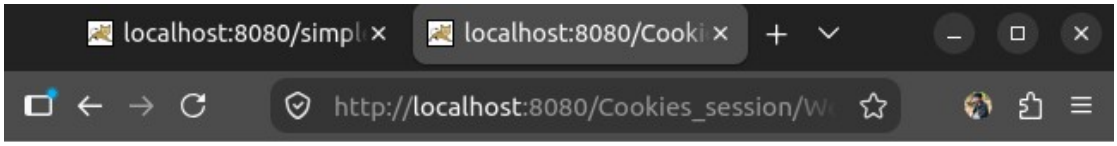
OUTPUT :



Login

Name: Ragupathi

Submit



Welcome, Ragupathi!

Exp. No: 14	FORWARDING REQUESTS AND REDIRECTING RESPONSES
Date:	

PROGRAM :

Homepage.html :

```
<html>
<head>
   <title>Age Verification</title>
</head>
<body>
<h2>Enter Your Details</h2>
<form action="NewServlet" method="post">
   Name:
   <input type="text" name="username" required><br><br>
   Age:
   <input type="number" name="age" required><br><br>
   <input type="submit" value="Submit">
</form>
</body>
</html>
```

Error.html :

```
<html>
<head>  <title>Error</title> </head>
<body>
<h2>You are not eligible to vote.</h2>
<p>You must be 18 years or older to vote.</p>
<a href="homepage.html">Go Back</a>
</body>
</html>
```

NewServlet.java :

```
import java.io.IOException;
import javax.servlet.RequestDispatcher;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/NewServlet")
public class NewServlet extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {
        String username = request.getParameter("username");
        int age = Integer.parseInt(request.getParameter("age"));
        if (age >= 18) {
            request.setAttribute("username", username);
            RequestDispatcher rd =
                    request.getRequestDispatcher("Welcome");
            rd.forward(request, response);
        } else {
            response.sendRedirect("error.html");
        }
    }
    @Override
```

```
protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
  }
  @Override
  protected void doPost(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
  }
}
```

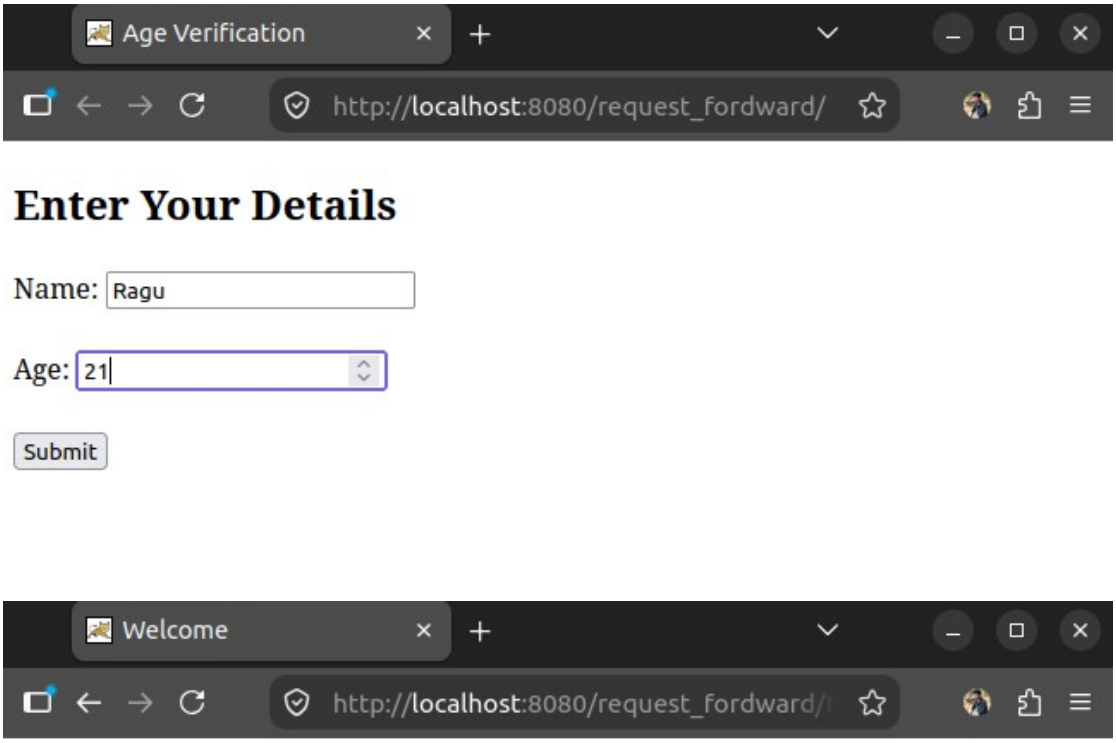
Welcome.java :

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/Welcome")
public class Welcome extends HttpServlet {
  protected void processRequest(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
```

```
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String username = (String) request.getAttribute("username");
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Welcome</title>");
        out.println("</head>");
        out.println("<body>");
        out.println("<h2>Welcome " + username + "!</h2>");
        out.println("<p>You are eligible to vote.</p>");
        out.println("</body>");
        out.println("</html>");
    }
    @Override
    protected void doGet(HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
}
    @Override
    protected void doPost(HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
}
```

OUTPUT :


Age Verification
http://localhost:8080/request_fordward/
Enter Your Details
Name: Ragu
Age: 21
Submit
Welcome
http://localhost:8080/request_fordward/
Welcome Ragu!
You are eligible to vote.

Exp. No: 15	WEB APPLICATIONS USING DATABASE
Date:	

PROGRAM :

Database code :

```
mysql> use ragu
Database changed
mysql> create table stud(user varchar(20),pass varchar(20));
Query OK, 0 rows affected (0.11 sec)

mysql> insert into stud values("ragu","ragu@123");
Query OK, 1 row affected (0.02 sec)

mysql> select*from stud;
+------+----------+
| user | pass     |
+------+----------+
| ragu | ragu@123 |
+------+----------+
1 row in set (0.00 sec)
```

Index.html :

```
<html>

<head>

   <title>Login Page</title>

</head>

<body>

<form action="LoginServlet" method="post">

   <h2>Login Form</h2>

   Username:

   <input type="text" name="username" required><br><br>

   Password:

   <input type="password" name="password" required><br><br>

   <input type="submit" value="Login">

</form>

</body>

</html>
```

DatabaseConnection.java :

```
import java.sql.Connection;
import java.sql.DriverManager;

public class DatabaseConnection {
   static Connection con;
   public static Connection initializeDatabase() {
      try {
         Class.forName("com.mysql.cj.jdbc.Driver");
         con = DriverManager.getConnection(
               "jdbc:mysql://localhost:3306/ragu",
               "root",
               "ragu@123");
      } catch (Exception e) {
         e.printStackTrace();
      }
      return con;
   }
}
```

LoginServlet.java :

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
```

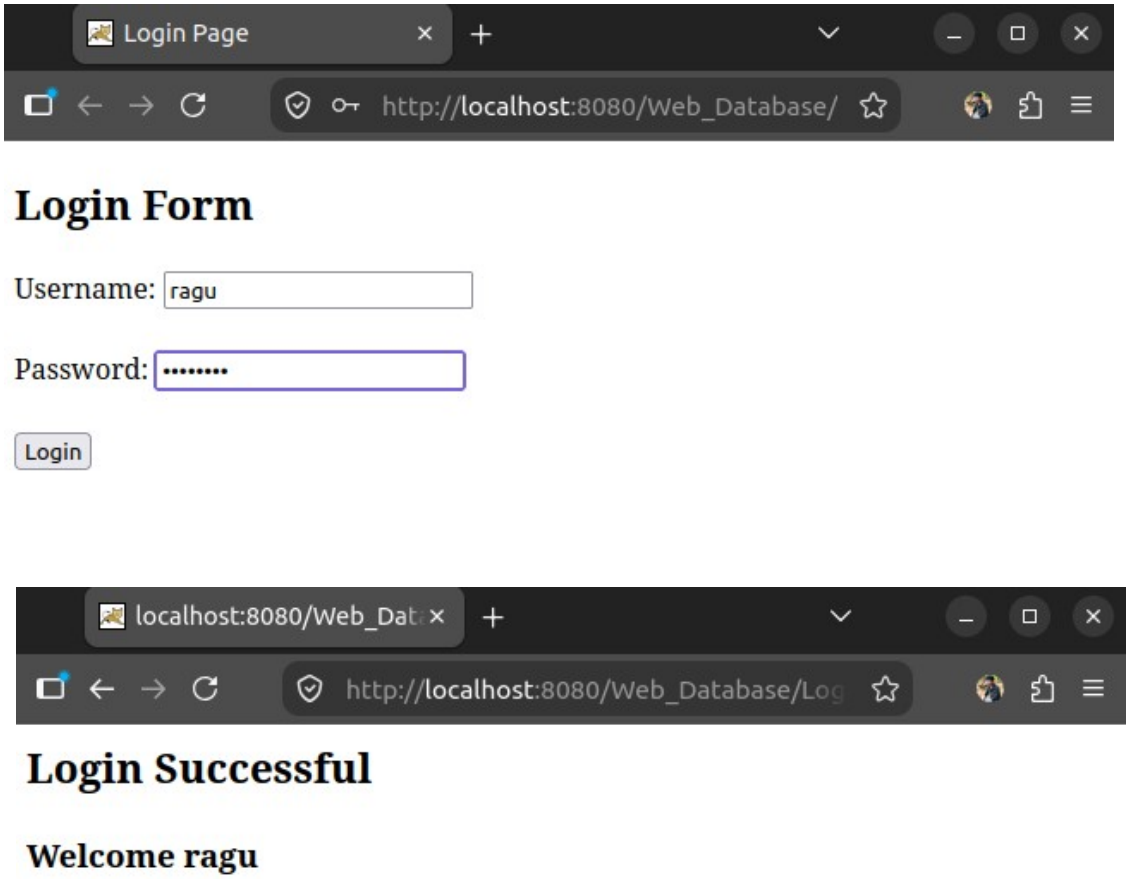
```
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {
   @Override

   protected void doPost(HttpServletRequest request,
         HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      String username = request.getParameter("username");
      String password = request.getParameter("password");
      try {
         Connection con = DatabaseConnection.initializeDatabase();
         String sql = "SELECT * FROM stud WHERE user=? AND pass=?";
         PreparedStatement ps = con.prepareStatement(sql);
         ps.setString(1, username);
         ps.setString(2, password);
         ResultSet rs = ps.executeQuery();
         if (rs.next()) {
            out.println("<html>");
            out.println("<body>");
```

```
                out.println("<h2>Login Successful</h2>");
                out.println("<h3>Welcome " + username + "</h3>");
                out.println("</body>");
                out.println("</html>");
            } else {
                out.println("<html>");
                out.println("<body>");
                out.println("<h2>Invalid Username or Password</h2>");
                out.println("</body>");
                out.println("</html>");
            }
            con.close();
        } catch (Exception e) {
            out.println("<h2>Database Error</h2>");
            e.printStackTrace(out);
        }
    }
}
```

OUTPUT :


Login Page
http://localhost:8080/Web_Database/
Login Form
Username: ragu
Password: ••••••••
Login

localhost:8080/Web_Dat
http://localhost:8080/Web_Database/Log
Login Successful
Welcome ragu

Exp. No: 16	DEVELOPING SIMPLE BEANS
Date:	

PROGRAM :

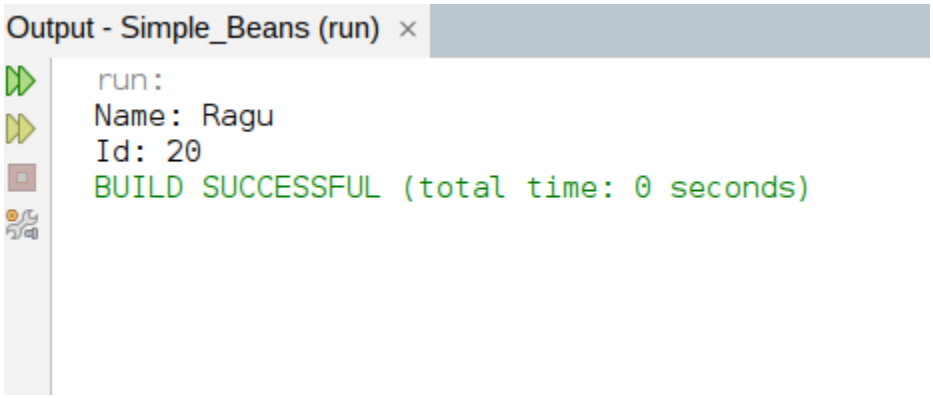
Student.java :

```
package stud;
public class Student {
    private String name;
    private int id;
    public Student() {
    }
    public Student(String name, int id) {
        this.name = name;
        this.id = id;
    }
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public int getId() {
        return id;
    }
    public void setId(int id) {
        this.id = id;
    }
}
```

TestStudent.java :

```
import stud.Student;
public class TestStudent {
   public static void main(String[] args) {
      Student s1 = new Student();
      s1.setName("Ragu");
      s1.setId(20);
      System.out.println("Name: " + s1.getName());
      System.out.println("Id: " + s1.getId());
   }
}
```

OUTPUT :


Output - Simple_Beans (run)
run:
Name: Ragu
Id: 20
BUILD SUCCESSFUL (total time: 0 seconds)

Exp. No: 17	USE BEANS WITH JSP TAGS
Date:	

PROGRAM :

Index.jsp :

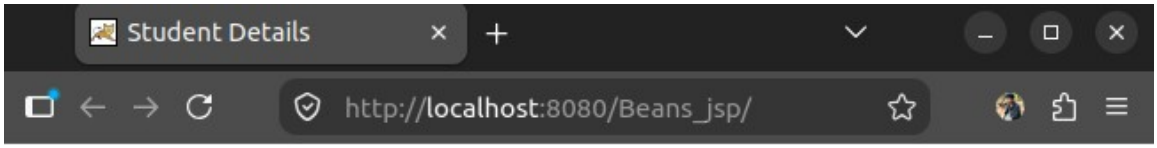
```
<%@ page import="student.Student" %>


<html>
<head>
    <title>Student Details</title>
    <style>
        h1{
            color: blue;
        }
    </style>
</head>
<body>
<h1>Student Details</h1>
<%
    Student student = new Student();
    student.setName("Ragupathi");
    student.setId(21);
    request.setAttribute("student", student);
%>
<p>Name : ${student.name}</p>
<p>ID : ${student.id}</p>
</body>
</html>
```

Student.java :

```
package student;
public class Student {
    private String name;
    private int id;
    public Student() {
    }
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public int getId() {
        return id;
    }
    public void setId(int id) {
        this.id = id;
    }
}
```

OUTPUT :



Student Details

Name : Ragupathi

ID : 21